

Product Requirements Document (PRD)

Wearable Proof-of-Human-Activity (PoHA) System

1. Product Purpose

The PoHA Wearable System replaces traditional energy-intensive cryptocurrency mining with verification of real human activity. The product uses wearable sensors to measure body and cognitive activity while users perform assigned tasks. Verified activity is used as proof-of-work and recorded on a blockchain.

2. Problem Statement

Traditional cryptocurrency mining requires massive computational power and energy. This system solves that problem by using unavoidable, real human biological signals as a secure and sustainable proof mechanism.

3. System Overview (Infographic Explanation)

Component	Role
Task Server	Assigns tasks such as CAPTCHA, content viewing, or interaction challenges
Wearable Sensor	Measures body activity (EEG, heart rate, heat, motion, brain activity)
User Device	Processes raw sensor data into encoded or hashed activity data
Communication Network	Securely transmits data between components
Cryptocurrency System	Verifies activity data and awards cryptocurrency
Blockchain Ledger	Records verified activity as immutable proof-of-work

4. Wearable Device & Sensors

The wearable device contains one or more sensors capable of detecting physical and cognitive body activity. These signals are difficult to fake and provide a strong human-authenticity signal.

Sensor Type	Measured Signal	Purpose
EEG / Brain Sensors	Brain waves (alpha, beta, gamma)	Detect focus, cognition, human thought
Heart Rate Monitor	Pulse, variability	Measure engagement and physical response
Thermal Sensor	Body heat radiation	Confirm live human presence
Motion / Optical Sensors	Movement, eye/facial motion	Verify task interaction

5. Functional Flow (User Journey)

1. User receives a task on their device 2. User performs the task naturally 3. Wearable sensor captures body activity 4. Device encodes or hashes activity data 5. Data is sent to the cryptocurrency system 6. System verifies authenticity and difficulty 7. User is rewarded and a block is recorded

6. Data Processing & Verification

Captured body activity is converted into symbolic data, vectors, or hashes. Verification checks include pattern matching, similarity scoring, difficulty targets, and re-hashing to ensure data integrity and prevent spoofing.

7. Blockchain & Rewards

Once verified, activity data becomes proof-of-work. A new block is generated containing the task, activity hash, reward amount, and previous block hash, ensuring immutability and transparency.

8. Security & Privacy Considerations

- Raw biometric data can remain on-device
- Only hashed or vectorized data is transmitted
- Re-hashing prevents replay attacks
- Human activity is statistically difficult to fake
- Supports compliance with privacy regulations

9. Success Metrics

- $\geq 99\%$ valid human activity verification
- < 2 seconds verification latency
- Reduced energy usage vs traditional mining
- Scalable to thousands of concurrent users

10. Conclusion

The PoHA Wearable System introduces a sustainable, human-centric alternative to cryptocurrency mining. By combining wearable technology, biometric sensing, and blockchain verification, the product enables secure proof-of-work derived directly from real human activity.